# CS3101 PROJECT REPORT
# ZIPRATHANS

**Athul Tom (21MS047)**
**Afrin K Riaz (21MS059)**
**Karthiayani Vijai (21MS127)**

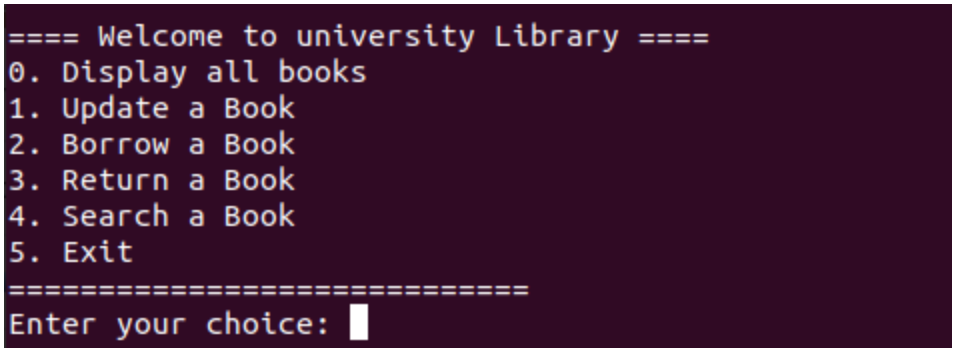
For this project we have designed a online library system. It was a collective effort by all the group members. We have used 4 files for this project.

1. “file.txt” for storing the book details
2. “users.txt” for storing the user details and passwords
3. “userissued.txt” for storing the number of books issued by the user
4. “bookissued.txt” for storing the information of books that are issued

The project was divided into modules. Everyone take part in the interface design. The functions to search books where written by Athul Tom (21MS047). He also write the code for the main() and the code for return_book(BOOK *book[1000], char* username). Afrin K Riaz (21MS059) has written codes for updating,deleting and inserting books. Karthiayani Vijai (21MS127), has written codes for issuing books. She created structures for storing details from file and written codes for scanning the values of file to structure.. Everyone helped each other in writing codes.

The code will only run on linux operating system as strcasecmp function is used.
When you run the code itr asks for the username and password.
The username is given in such a way that Admins username starts with ‘A’, Student starts with ‘S’, faculty with ‘F’.
The format of the username is like A0001 and password is like Ap0001.

After successful login this will appear in terminal




Same menu will be shown to both users as well as the admin, but access to functions is separate for them. A user will not be allowed to update a book, only the admin will be able to do it.
When the user asks for searching a book it gives several options allowing the user to search by a variety of ways including searching for the author or publisher.

If the user is intending to issue a book he/she is expected to know the ID of the book. To get the ID of the book he/she can use the help of a search facility. Different book id,title,author,publisher and number of copies are given in file “file.txt” in TAB separated manner.
Example of a book id = CH0001  (Book name = Physical Organic Chemistry).

When the book is issued the code intends to update the database of the number of books of that particular book remaining in the library. That is, the number of copies of the book gets reduced.
When the book is returned the number of copies of the book available will increase. Here only a single element of the structure books is changed(No.of books).
In the case of Updating books multiple elements of a structure gets changed.

Issue function includes a series of statements which checks specific cases.In this function after checking out multiple conditions inside a loop, data is copied into a temporary file, such that a change in a given structure is also included in the temporary file. After exiting from the loop the original files data are getting deleted and contents of the newly made temporary file is copied into the original file. Thus by doing this changes are actually reflected in the original file.
Similar kind of approach is made for book_return function.